MORPHE Architecture & Systems Engineering Log

User: Let's design the ingestion pipeline for unstructured engineering docs.

Assistant: For ingestion, we will implement FastCDC content-defined chunking paired with LayoutLM to preserve document tables, headings, and semantic hierarchies.

User: How should we structure the in-memory graph storage for microsecond traversals?

Assistant: We will build a multi-threaded C++17 CAS core utilizing lock-free concurrent adjacency lists and Zstandard dictionary compression for storing graph diffs.

User: What database are we using for storing initial module metadata?

Assistant: Initially, we planned to store module metadata directly in SQLite tables with WAL mode enabled.

User: Wait, update on the database requirement: we are deprecating SQLite as active storage. We replaced SQLite with LMDB in-memory key-value store for zero-copy lookups. SQLite is now strictly used as cold persistent ledger backup.

Assistant: Confirmed update: We have superseded SQLite with LMDB as our active primary in-memory engine. SQLite is now relegated strictly to cold immutable ledger backup storage.

User: What algorithm will drive the autonomous topological reconfiguration?

Assistant: Autonomous restructuring is driven by a LinUCB contextual bandit coupled with Proximal Policy Optimization (PPO). The policy evaluates subgraphs encoded via Graph Neural Networks (GNN).

User: How does the hybrid retrieval engine route queries?

Assistant: Queries run in parallel across FAISS dense HNSW vectors and BM25 sparse lexical indexes, followed by Personalized PageRank graph expansion across adjacent nodes, and final Cross-Encoder re-ranking.

User: How do we prevent catastrophic regressions when the bandit proposes a mutation?

Assistant: Every mutation must pass a 7-stage gate: off-policy IPS pre-screening, isolated Copy-on-Write (CoW) sandbox trial, verifiable mathematical reward calculation (Recall, MRR, Latency, Traversal Cost), and bootstrapped confidence bounds before committing to the immutable ledger.

User: What happens if a human operator manually locks a specific knowledge path?

Assistant: When is_human_locked is flagged on a node or edge, it creates an absolute mask against the GNN state. The PPO bandit is mathematically forbidden from mutating or pruning that pathway.